

Before entering sensitive information:

1. Inspect the address bar and confirm the correct domain.
2. Confirm that the connection uses [https://](#).
3. Never bypass a certificate warning merely because the page looks familiar.
4. Avoid following login links from messages; type the known address or use a trusted bookmark.

Use HTTPS regardless of whether you also use a VPN or Tor. A tunnel protects one segment of the route. HTTPS continues protecting traffic between the tunnel's exit point and the website.

Private DNS: It hides lookups, not browsing

The Domain Name System translates names such as a website address into IP addresses. Traditional DNS requests may travel without encryption, allowing someone on the local network or the internet provider to observe which domains are being looked up.

DNS over HTTPS, or DoH, sends these requests through an encrypted HTTPS connection to a compatible resolver. Mozilla says this can hide domain lookups from public Wi-Fi observers, the ISP and other parties on the local network. However, the chosen resolver can receive the queries, and the website or its IP address may remain visible through other parts of the connection.

Private DNS should therefore be viewed as resolver privacy, not anonymity. It can also interfere with parental controls, organisational filtering or malware-blocking systems that rely on local DNS.

Configure DoH in Firefox

1. Open Settings > Privacy & Security.
2. Find DNS over HTTPS.
3. Choose a protection level.
4. Select the default resolver or a provider you understand.
5. Use a stronger no-fallback mode only after checking that required work, school or home filtering still functions.

Android's system-wide Private DNS generally uses DNS over TLS rather than DoH. Search Settings for "Private DNS", choose a provider hostname from a service you trust and test connectivity. Do not copy an unknown hostname from a social-media post.

Proxies change one application's path

A proxy receives requests from an application and sends them onward. The destination may see the proxy's IP address instead of yours. The limitation is scope. A browser proxy may not carry traffic from email, cloud-sync, update services or other applications. Some proxy types do not encrypt the connection between you and the proxy. A free web proxy can read pages, inject advertising, log destinations or alter unencrypted traffic.

Treat a proxy as a routing tool, not an anonymity product. Ask whether it supports HTTPS tunnelling, who operates it,

what it records and which applications actually use it. Never enter passwords through an unknown proxy.

A VPN moves trust from ISP to provider

A commercial VPN creates an encrypted tunnel from your device to a VPN server. The local Wi-Fi operator and ISP generally see the VPN connection rather than each destination reached through it. Websites see the VPN server's public IP address, which can reduce direct IP-based location and make many subscribers appear to share an address.

The VPN provider occupies a powerful position. It may know your originating IP address, connection times, assigned server and account details, and it can potentially observe destination information. EFF describes a VPN as a way to choose a different network path, not a complete defence against malware, tracking or government surveillance.



A VPN does not stop a signed-in platform from knowing who you are. It does not remove cookies, browser fingerprints, phone numbers, payment trails or information submitted in forms. It cannot make unlawful conduct lawful.

Evaluate a VPN before paying

Use this checklist:

1. **Purpose:** Decide whether you need protection on untrusted networks, an alternate IP location, workplace access or broader identity separation.
2. **Operator:** Identify the legal company, owners and country of operation.
3. **Logging:** Look for precise statements about connection, activity, device and account logs. "No browsing logs" may still permit extensive connection records.
4. **Evidence:** Prefer independently published audits or documented legal cases over influencer endorsements.
5. **Applications:** Confirm support for your operating systems and whether all traffic is routed.
6. **Leak controls:** Look for DNS handling, IPv6 support and a kill switch.
7. **Business model:** Ask how a free provider pays for servers, bandwidth and support.